

Deepak Paragi

AI & Machine Learning Engineer — Full Stack Developer
Bengaluru, India
deepakparagi03@gmail.com — +91 8197174493
LinkedIn — GitHub — Portfolio

Professional Summary

AI & Machine Learning Engineering student specializing in building production-grade AI systems, scalable backend architectures, and intelligent automation pipelines. Strong expertise in Retrieval-Augmented Generation (RAG), LLM integrations, and full-stack development. Proven ability to design modular systems, optimize performance, and deploy real-world applications with measurable impact.

Technical Skills

AI / ML: LLMs, RAG Pipelines, Embeddings, Vector Databases, Prompt Engineering, LangChain

Backend: Node.js, Express, REST APIs, Authentication, Webhooks, Middleware

Frontend: React.js, Tailwind CSS

Languages: Python, JavaScript, SQL, C++

Databases: MongoDB, SQL (Joins, Indexing, Optimization)

Core CS: Data Structures, Algorithms, OOP, DBMS, OS, Computer Networks

Tools: Git, GitHub, Postman, Vercel, Docker (basic), Jupyter

Professional Experience

Full Stack Developer Intern

GTTC, Hubli

Feb 2023 – Jun 2023

- Built and deployed a responsive web application improving usability and frontend performance
- Designed RESTful APIs enabling efficient client-server communication with reduced latency
- Optimized SQL queries and database schema improving data retrieval performance by 30%+
- Ensured API reliability through testing, validation, and debugging using Postman
- Collaborated in an agile environment using Git workflows for feature development and deployment

Projects

Enterprise Semantic Search Engine (RAG)

Python, LangChain, Pinecone, React

- Designed and implemented a scalable Retrieval-Augmented Generation system for querying large unstructured datasets
- Built document ingestion pipeline including chunking, embedding generation, and vector storage
- Engineered semantic retrieval layer using vector similarity search for context-aware responses
- Developed prompt orchestration system to inject relevant context, improving response accuracy
- Architected modular pipeline supporting extensibility and multi-source data integration

Automated AI Code Review Agent

Node.js, GitHub Actions, LLM APIs

- Built an AI-powered system to analyze pull request diffs and generate contextual review feedback
- Implemented diff parsing engine to structure code changes for LLM processing
- Designed prompt pipelines to detect bugs, security risks, and performance issues
- Integrated GitHub webhooks and CI/CD workflows for real-time automation
- Reduced manual review effort and improved code quality consistency across repositories

AI-Powered Fitness & Biomechanics Tracker

JavaScript, Python, MediaPipe

- Developed full-stack application generating personalized workout plans based on user data
- Integrated computer vision models to analyze posture and movement in real time
- Built feedback system detecting incorrect form and providing corrective suggestions
- Designed backend APIs for user data processing and performance tracking
- Optimized for low-latency interaction and responsive UI experience

Multi-Agent Data Automation Orchestrator

Python, Pandas, LLM APIs

- Designed automated data pipeline for cleaning, transforming, and analyzing datasets
- Implemented preprocessing modules for handling missing values and normalization
- Built multi-agent workflow for generating insights, summaries, and anomaly detection
- Developed end-to-end system from raw data ingestion to automated reporting
- Reduced manual data analysis effort by 85%

AI Systems & Architecture

- Designed RAG pipelines including ingestion, embedding, retrieval, and response generation layers
- Implemented prompt engineering strategies to improve output quality and reduce hallucinations
- Built LLM orchestration pipelines with structured inputs and controlled outputs
- Optimized vector search retrieval for relevance and performance

Backend Engineering & Scalability

- Designed modular REST API architectures with middleware and routing layers
- Implemented asynchronous processing for efficient request handling
- Built event-driven systems using webhooks and trigger-based workflows
- Applied query optimization and basic caching strategies for performance improvements

Data Engineering

- Built ETL pipelines for structured and unstructured datasets using Python
- Automated preprocessing using Pandas for transformation and cleaning
- Designed workflows for data analysis and insight generation
- Implemented anomaly detection and automated reporting pipelines

Engineering Practices

- Version control using Git with structured branching and collaboration workflows
- API testing, debugging, and validation using Postman
- Deployment using Vercel and cloud platforms
- Writing modular, scalable, and maintainable production-ready code

Education

Bachelor of Engineering – Artificial Intelligence & Machine Learning NMIT, Bengaluru	2023 – 2026
Diploma in Computer Science and Engineering Government Polytechnic, Gadag	2020 – 2023

Certifications

- Python Programming — Udemy
- Database Management Systems — Infosys Springboard
- Introduction to Machine Learning — Great Learning